

Wheat Disease Monitoring in Punjab, Pakistan

An Overview of Spot Blotch and Wheat Blast Surveillance

Punjab, Pakistan, has long been a focal point for wheat disease surveillance due to its extensive wheat production and diverse agro-ecological zones. Recent studies have concentrated on two major foliar pathogens: spot blotch caused by *Bipolaris sorokiniana* and wheat blast caused by *Magnaporthe oryzae* Triticum pathotype. This report summarizes key findings from surveys conducted between 2005 and 2020, highlighting incidence trends, disease severity, and the current status of wheat blast in the region.

Spot Blotch Surveillance

Surveys carried out in 2005, 2006, and 2008 across eight agro-ecological zones identified zone 6 (encompassing much of Punjab) as the hotspot for spot blotch, recording an incidence of 84 % and a disease index of 26.8. Severity scores reached 2.5 on the standard scale. In 2009, a comprehensive assessment found spot blotch to be the most prevalent disease, affecting 100 % of the Bhakkhar-2001 wheat variety with a disease index of 21.8 %. These data underscore the persistent threat posed by spot blotch to wheat yields in Punjab.

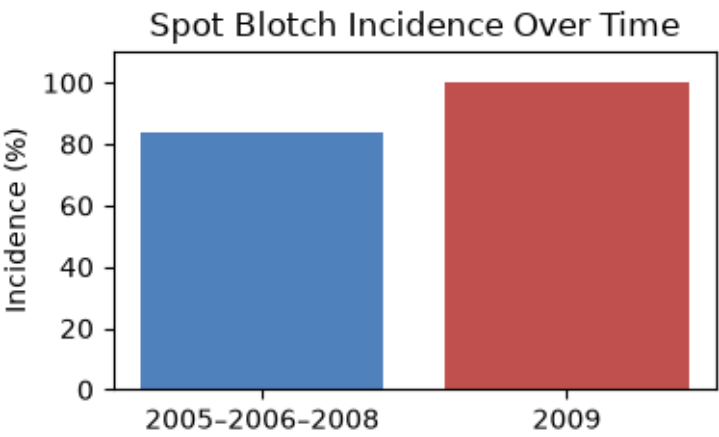
Wheat Blast Surveillance

During the 2018–2019 and 2019–2020 crop seasons, extensive field surveys and seed sample analyses across the province detected no presence of wheat blast. This indicates that the pathogen has not yet established itself in Punjab, although continued vigilance is essential given its rapid spread in other South Asian regions.

Data Comparison Table

Year	Zone/Variety	Incidence (%)	Disease Index	Severity Score
2005–2006–2008	Zone 6	84	26.8	2.5
2009	Bhakkhar-2001	100	21.8	—

Incidence Trend Chart



References

- [academia.edu](https://www.academia.edu)
- [researchgate.net](https://www.researchgate.net)
- [semanticscholar.org](https://www.semanticscholar.org)
- [sabrajournal.org](https://www.sabrajournal.org)
- [pakbs.org](https://www.pakbs.org)
- [cabidigitallibrary.org](https://www.cabidigitallibrary.org)